

Utkarsh Mishra

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 Github  LinkedIn  Website  Scholar

Education

- **Texas A&M University (TAMU)** 2025 – 2027
M.S. Computer Engineering
CGPA: 4.00/4.00
- **Indian Institute of Technology Gandhinagar (IIT-GN)** 2020 – 2024
B.Tech in Electrical Engineering with Minor in Computer Science
CGPA: 8.79/10

Publications

- [1] City Navigation in the Wild: Exploring Emergent Navigation from Web-Scale Knowledge in MLLMs (Paper)
*Utkarsh Mishra**, *Dwip Dalal**, *Narendra Ahuja*, *Nebojsa Jojic* (Project Page)
European Chapter of the Association for Computational Linguistics (EACL '26) [ORAL PRESENTATION]
- [2] Constructive Distortion: Improving MLLMs with Attention-Guided Image Warping (Paper)(Project Page)
Dwip Dalal, *Gautam Vashishtha*, *Utkarsh Mishra*, ... *Svetlana Lazebnik*, *Heng Ji*, *Unnat Jain*
International Conference on Learning Representations (ICLR'26)
- [3] Transformer Augmented Multi-Resolution Hash Encoding in Diffusion Model for 3D Point Cloud Denoising (Paper)
Seema Kumari, *Utkarsh Mishra*, *Srimanta Mandal*, *Shanmuganathan Raman*
International Conference on Image Processing (ICIP'25)

*Equal contribution

Work Experience

- **Research Engineer @ Department of AGLS, Texas A&M University** (Oct 2025 – Present)
 - Leading technical development for Project SMART, a National Institute of Health (NIH) funded research study on adolescent smartphone behavior.
 - Designing a scalable machine learning pipeline (ETL) for over 1M screenshots using MLLM-based usage pattern and sentiment analysis and automated data annotation; deploying batch inference on GCP / HPC clusters. Managing Android app development and deployment (Stanford Screenomics).
- **System Analyst @ HDFC Bank Ltd** (Jul 2024 – Aug 2025)
 - Member of the CMDB team responsible for Discovery and onboarding of organizational IT infrastructure to ServiceNow. Developed automation solutions for identifying and troubleshooting server Discovery errors.
- **System Analyst Intern @ HDFC Bank Ltd** (May 2023 – Jul 2023)
 - Member of the CMDB team that facilitated the transition from CA ServiceDesk R17 to ServiceNow. Developed automation tools to accelerate troubleshooting. **Received a full-time offer based on excellent performance.**

Research Experience

- **Research Collaborator @ University of Illinois, Urbana-Champaign** (Jan 2025 – Present)
Collaborators: Prof. Svetlana Lazebnik, Prof. Unnat Jain, Dwip Dalal (PhD Student)
 - **Constructive Distortion: Improving MLLMs with Attention-Guided Image Warping** (Project Page)
Designed an image warping algorithm “Attwarp” that increases semantically relevant content of an image using an MLLM’s own query conditioned attention map over the image. Achieved **SOTA** performance on LLaVA and Qwen-VL on TextVQA, GQA, MMMU, POPE, and DocVQA datasets. [2]
 - **City Navigation in the Wild: Exploring Emergent Navigation from Web-Scale Knowledge in MLLMs** (Project Page)
Constructed an environment to evaluate MLLMs on long-horizon, sparsely grounded sequential decision-making tasks. Empirically demonstrated the limitations of reasoning methods (GEPA, Reflection, etc) in this setting, and proposed the Verbalization of Path (VoP) framework, which elicits latent world knowledge from MLLMs. VoP attains state-of-the-art performance on instruction-free, goal-driven outdoor navigation tasks. [1]
- **Research Assistant @ IIT Gandhinagar** (Aug 2023 – Feb 2025)
Mentor: Prof. Shanmuganathan Raman, Prof. Ravi Hegde
 - **Diffusion Transformer for 3D Point Cloud Denoising**
Developed a transformer-based model for point cloud denoising building upon P2P-Bridge method by using multi-resolution hash encoding, providing an alternate to PointNet based networks. Achieved SOTA performance on PCNet and PUNet datasets at 1% noise levels. [3]
 - **Denoising Gaussian Splatting For 3D Scene Reconstruction** (Repository)
Extended 3D Gaussian Splatting by applying denoising techniques (DBSCAN, point-wise distance pairing) to the input COLMAP sparse point cloud to reduce visual artifacts and developed novel regularization methods.
 - **Human Pose Classification using Spatial-Temporal Graph Neural Network** (Poster)(Repository)
Developed a graph-neural-network-based human-pose classifier. Utilized OpenPose, AlphaPose, and YOLOv7 for keypoint detection and pose-graph construction. Implemented a hybrid Graph Convolutional Network (GCN) – LSTM architecture. Trained and evaluated the model on the MPOSE-2021 dataset.

Projects

- **VGAP: Vision-Gated Action Planning for Web Agents** (Aug 2025 – Dec 2025)
CSCE 642 Deep Reinforcement Learning, Texas A&M University ([Repository](#))
 - Developed VGAP, a visual grounding module for autonomous web agents that uses Direct Preference Optimization (DPO) to train a lightweight Qwen2-0.5B model for predicting optimal screenshot crop regions, reducing input complexity for downstream vision-language action models.
 - Implemented parameter-efficient fine-tuning using LoRA adapters via Hugging Face PEFT and TRL libraries; generated 6,000+ DPO preference pairs from the Mind2Web multimodal dataset containing human-annotated web navigation trajectories.
 - Integrated VGAP into the SeeAct web automation framework with LiteLLM for universal VLM support (GPT-4o, Claude, Gemini, LLaVA); implemented robust inference pipeline with bounding-box extraction, multi-strategy coordinate parsing, and graceful fallback mechanisms.
- **Analyzing Physics Understanding in Video Generation Models** (Aug 2025 – Dec 2025)
CSCE 689 Vision Foundation Models, Texas A&M University ([Project Page](#))([Repository](#))
 - Evaluated whether general-purpose video generation models (Sora, Veo3, Grok, Wan2.1) can produce physically realistic data for robotics training by adapting the RIGVid (CVPR'25) pipeline for 6D pose estimation from generated videos.
 - Designed physics benchmark suite testing model adherence to gravity, buoyancy, and collision dynamics; built kinematics analysis pipeline using SAM2 for video object segmentation, centroid tracking, and numerical differentiation with Savitzky-Golay smoothing.
- **Synthetic Data Generation for Machine Learning** (Aug 2023 – May 2024)
Mentor: Prof. Shanmuganathan Raman, IIT Gandhinagar ([Poster](#))([Repository](#))
 - Generated high-quality synthetic images using StyleGAN-XL and Stable Diffusion tailored to the CIFAR-10 dataset, producing over 90,000 images to augment training data.
 - Engineered advanced prompts by extracting visual attributes from the Visual Genome dataset, aligning class labels with WordNet synsets, clustering visual attributes with GloVe embeddings, and sampling from clusters to increase diversity of attributes. Further prompt enhancements using Gemma 2B-it.
 - Conducted comparative analysis of classifiers (e.g. ResNet-34) trained on varying ratios of synthetic to real data through extensive experimentation on different types of synthetic data. Analyzed effects on classifier accuracy and confusion matrix.
- **Japanese Language Learning App** (May 2024 – Jun 2024)
Personal Project ([Repository](#))
 - Developed a native Android application in Kotlin using Jetpack Compose with an integrated open-source JMDICT dictionary to enhance Japanese language learning by providing contextual word usage from YouTube videos.
 - Implemented natural language processing using Kuromoji to handle Japanese word conjugations by converting words into base forms, improving search accuracy and usability. Utilized the YouTube Transcript API with Python integration via Chaquopy for transcript processing.
- **GAN Inversion for Latent Space Analysis** (Jan 2024 – May 2024)
ES 413 Deep Learning, IIT Gandhinagar ([Repository](#))
 - Used GAN inversion on StyleGAN to invert images of cars into their latent representation and analyzed effect of object rotation on latent representation and generated novel views.
 - Identified and analyzed issues such as imperfect reconstruction, editability vs reconstruction error trade-off, the influence of the GAN's training data on outputs, background effects, and visual artifacts from latent manipulation.
- **Wearable Device for Real Time Sign Language Recognition** (Apr 2023)
ES 333 Microprocessors and Embedded Systems, IIT Gandhinagar ([Repository](#))
 - Worked in a team of 4 to develop a wearable device for sign language recognition using a STM32 Nucleo micro-controller and flex sensors for data acquisition.
 - Implemented USB CDC protocol to communicate sensor data to host device running Scikit-learn based MLP classifier used to detect gestures.

Technical Skills Summary

- **Languages:** Python, C/C++, Java, Kotlin, MATLAB, Verilog, Bash
- **Frameworks & Libraries:** PyTorch, TensorFlow/Keras, Hugging Face (Transformers, PEFT, TRL, Diffusers), LangChain, OpenAI API, OpenCV, Scikit-learn, Pandas, NumPy, Matplotlib
- **Tools & Platforms:** Linux, Git, Google Cloud, Firebase, Weights & Biases, ComfyUI, Docker, Jupyter, Colab
- **Technologies and Architectures :** Machine Learning, Artificial Intelligence, Deep Learning, Computer Vision, Natural Language Processing, LLMs, Vision-Language Models, LoRA/QLoRA, DPO, RLHF, Diffusion Models, GANs, VAEs, Transformers, Vision Transformers (ViT), CLIP, SAM, 3D Gaussian Splatting, NeRFs, Graph Neural Networks
- **Relevant Coursework:** Deep Learning, Deep Reinforcement Learning, Vision Foundation Models, Data Mining & Analysis, Machine Learning Engineering, Scientific Machine Learning

Interest Areas

Large Language Models (LLMs), Vision-Language Models (VLMs), Multimodal Learning, Agentic AI Systems, Reinforcement Learning from Human Feedback (RLHF), Diffusion Models, 3D Vision, Embodied AI, Synthetic Data Generation